

CPSI 28003 - 01 and 9S1 Algorithms

Fall 2025

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Homework 1 (No Late Homework)
Due on **9/18/2025** (Thursday) at the beginning of the class
Please submit Homework 1 to Blackboard
Mastering C Pointers

Total Pages: 8

Total Points: 100

Multiple Choice & Short Answers

1. (5 Points) If a variable is a pointer to a structure, then which of the following operator is used to access data members of the structure through the pointer variable?

a. $\rightarrow$
b. .
c. &
d. *

2. (5 Points) If a variable is a pointer to a float, then which of the following operator is used to access data through the pointer variable?

a. $\rightarrow$
b. .
c. &
d. *

3. (5 Points) How many bytes does an integer take in C/C++?

a. 2
b. 4
c. 6
d. 8

4. (5 Points) When variables c1 and c2 are declared continuously, are they allocated in memory continuously? Run the following program on your computer and print out the memory locations that are assigned to all the variables by your compiler. What are the memory locations of c1 and c2? Are the memory locations located next to each other?

char c1, c2;
[ANSWER]

The memory addresses for c1 is 00000039464FF654 and for c2 is 00000039464FF674. Yes they are next to each other in memory locations.

5. (15 Points) Given the following program, trace the output by **drawing** the memory model as the program runs. Assuming each cell is one byte memory.

5a. (5 Points)

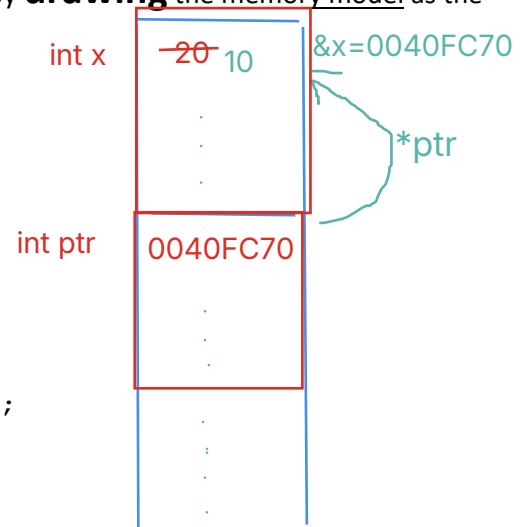
```
#include <iostream>
using namespace std;
int main()
{
    int x = 20;
    int *ptr;
    ptr = &x;
    *ptr = 10;
    cout << &x << " " << ptr << " " << endl;
    cout << x << " " << *ptr << endl;
    system("PAUSE");
    return 0;
}
```

output

0040FC70 0040FC70

10 10

Press any key to continue . . .



[ANSWER]

- 5b. (5 Points) Explain why the following statement in Sample 1

```
cout << x << " " << *ptr << endl;
prints 30 30 to the output?
```

// Sample 1

```
#include <iostream>
using namespace std;
```

```
void some_function(int *somePtr)
{
    int someValue = 30;
    *somePtr = someValue;
}
```

```
int main()
{
    int x = 20;
    int *ptr;
    ptr = &x;
    some_function(ptr);
    cout << x << " " << *ptr << endl; // explain how to print 30 30?
    system("PAUSE");
    return 0;
}
```

Output

30 30

Press any key to continue . . .

[ANSWER]

The function changes the value at the pointer's memory address to 30. Since the variable x's memory address is stored in ptr, changing the value of ptr also changes the value of x.

5c. (5 Points) Explain why the following statement in Sample 2

```
cout << x << " " << *ptr << endl;
```

prints 20 20 to the output? Why not 30 30?

// Sample 2

```
#include <iostream>
using namespace std;

void some_function(int *somePtr)
{
    int someValue = 30;
    somePtr = &someValue;
}

int main()
{
    int x = 20;
    int *ptr;
    ptr = &x;
    some_function(ptr);
    cout << x << " " << *ptr << endl;
    system("PAUSE");
    return 0;
}
```

Output

20 20

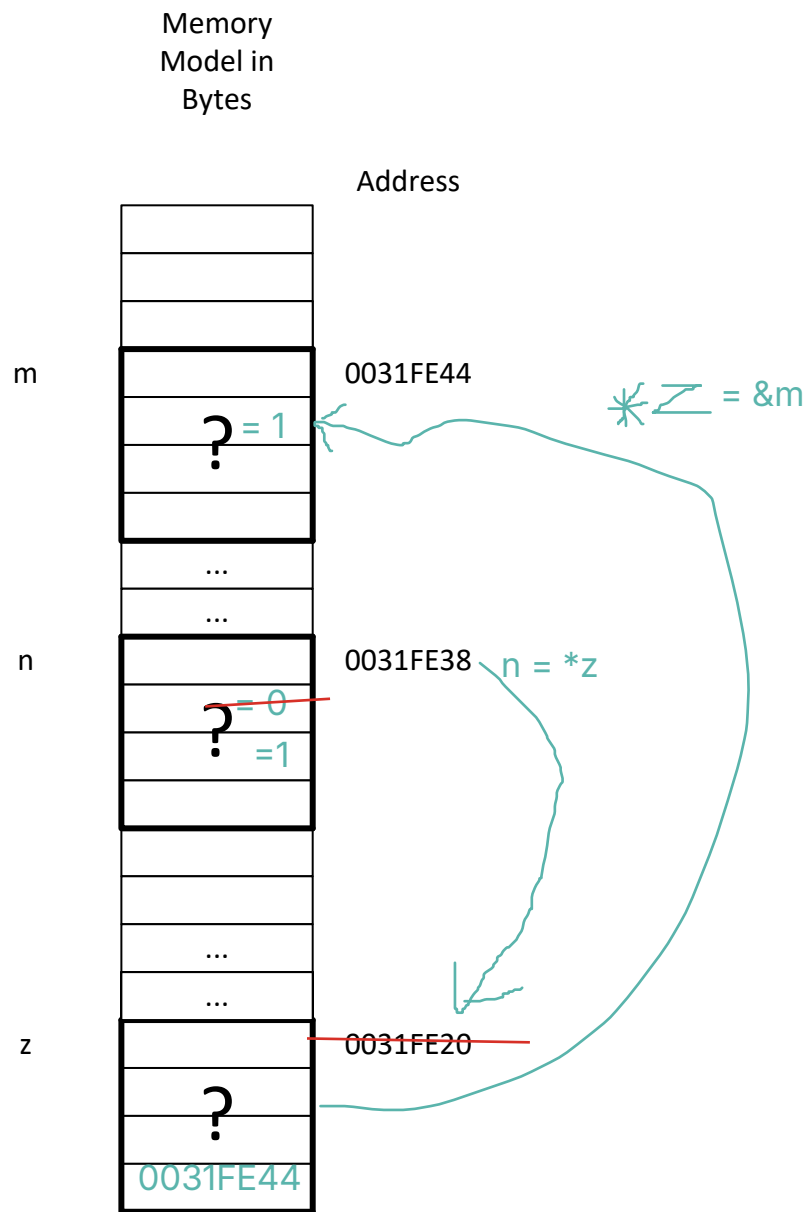
Press any key to continue . . .

[ANSWER]

The value of the ptr passed to some_function() is changed to the memory address of someValue, but since they just changed the memory address of ptr and not the value at the memory address of ptr, once the function goes out of scope, the variable someValue goes out of scope. Therefore, the variable x and *ptr stay 20.

6. (10 Points) Given the following program, trace the output by drawing the memory model as the program runs. Assuming each cell is one byte memory. Fill in the blanks.

```
#include <iostream>
using namespace std;
void main(void)
{
    int m = 1;
    int n;
    int *z = &m;
    n = *z;
    system("PAUSE");
}
```



[ANSWER]

7. (10 Points) Given the following program, (a) trace the output by drawing the memory model as the program runs. (b) What does `printf("%d\n", ptr2 - ptr1)` print in C/C++?

```
#include <iostream>
using namespace std;

int main()
{
    float arr[3] = {1.5, 1.0, 3.5};
    float *ptr1;
    float *ptr2;

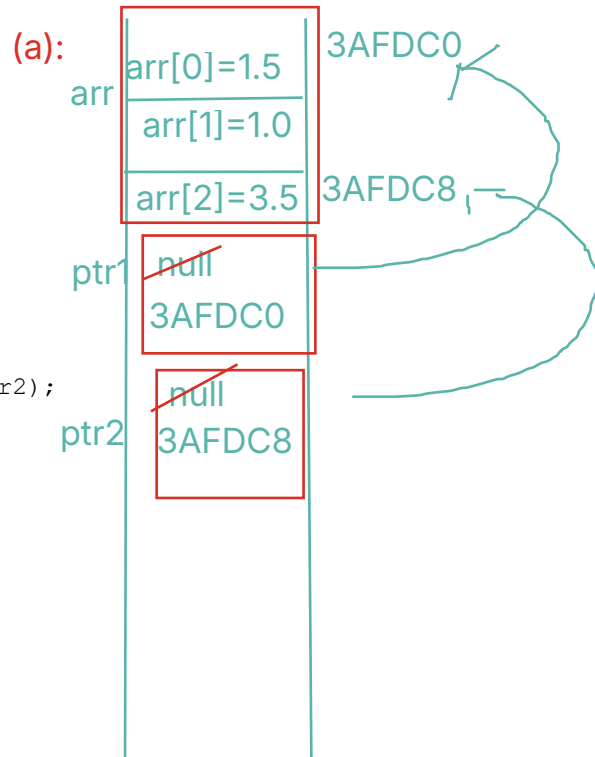
    ptr1 = &arr[0];
    ptr2 = ptr1 + 2;

    printf("%X %X %X\n", arr, ptr1, ptr2);

    system("PAUSE");
    return 0;
}
```

output
3AFDC0 3AFDC0 3AFDC8
Press any key to continue ...

[ANSWER] (b): 2



8. (6 Points) Given the following program,

// Sample

```
#include <iostream>
using namespace std;

#include <stdio.h>
int anArray[3] = {1, 2, 3};
int *pArray = anArray; // Statement 5

for (int i=0; i<3; i++)
{
    printf("%d", *pArray); // c style
    pArray++;
}

system("PAUSE");
return 0;
}
```

T (T/F) (2 Points) `pArray` is equivalent to `&anArray[0]` in Statement 5

T (T/F) (2 Points) In the for loop, `printf("%d", *pArray);` and `pArray++;` can be replaced by `printf("%d", *(pArray+i));`

T (T/F) (2 Points) *(pArray+2) is equivalent to &anArray[3]

9. (4 Points) Given the following program,

// Sample 9

```
#include <iostream>
using namespace std;
const int SIZE = 5;
int main()
{
    int anArray[SIZE] = {1, 2, 3};
    int *pArray = &anArray[0];

    int *ptr = pArray+1;
    int * ptr1 = ptr + 2;
    cout << *ptr << " " << *ptr1 << endl;

    system("PAUSE");
    return 0;
}
```

 T (T/F) (2 Points) number of elements between ptr and ptr1 is 2

 T (T/F) (2 Points) number of bytes between ptr and ptr1 is 8

10. (5 Points) What will be the output of the following code?

```
#include <iostream>
using namespace std;
void main()
{
    char *s = "hello";
    char *p = s;
    cout << *(p+3) << " " << s[1] << endl;
}
```

[ANSWER]

compiler error: a value of type "const char *" cannot be used to initialize an entity of type "char *"

11. (6 Points) Given the following code,

```
int i = 3, j = 5, *p = &i, *q = &j, *r;
float x;
```

Individual statements are independent. If compile error, just say so.

11.a (2 Pints) What is the output value? compiler error p == i;

11.b (2 Points) Is it legal? no. compiler error r = &x;

11.c (2 Points) What is the output value? 11 $7 * *p / *q + 7$

12. (5 Points) Show the output of the following C program?

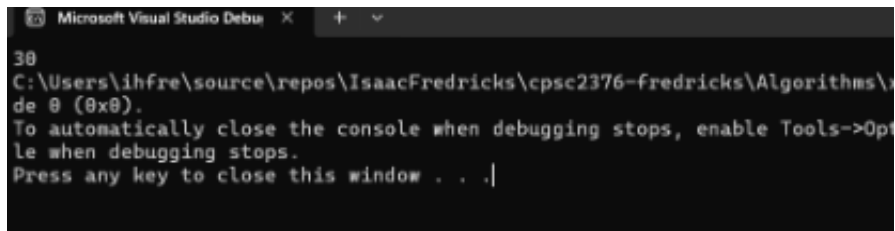
```
# include <stdio.h>
```

```
void xyz(int *ptr)
{
    *ptr = 30;
}
```

```
int main()
{
    int y = 20;
    xyz(&y);
    printf("%d", y);
}
```

```
return 0;
```

[ANSWER]



13. (19 Points) Write a short C program that declares and initializes (to any value you like) a Boolean. Next declare and initialize a pointer to the address of the Boolean variable. Your program then prints the address of, and value stored in, and the memory size (in bytes) of this Boolean variable using the pointer.

13a. (10 Points) Show the program.

[ANSWER]

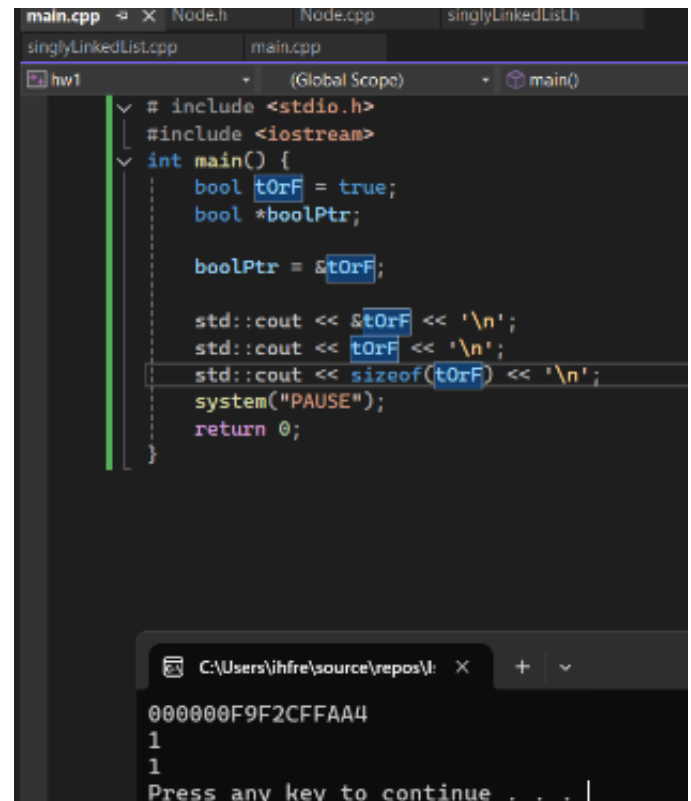
```
#include <iostream>
using namespace std;

int main()
{
    bool b = 100;
    bool *pB;

    pB = &b;

    printf("%p \n", &b);
    printf("%d \n", b);
    printf("%d \n", sizeof(b));

    system("PAUSE");
    return 0;
}
```



13b. (9 Points)

The address of Boolean 000000F9F2CFFAA4

The value of Boolean 1

The number of bytes allocated for Boolean variable 1

[ANSWER]